

MORNINGTON ISLAND, AustraliaWMO: **942560**Lat: **16.664S**Lon: **139.184E**Elev: **9**StdP: **101.22**Time Zone: **10.00 (E10)**Period: **94-13**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	15.5	3.5	4.9	23.5	5.2	5.5	22.2	9.2	18.4	8.5	19.1	1.1	140	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	5.6	34.4	23.9	33.6	24.6	33.0	25.1	28.5	31.3	28.0	30.8	27.5	30.5	3.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.7	23.7	30.0	27.2	23.0	29.8	26.7	22.3	29.4	92.1	31.3	89.8	30.8	87.6	30.6	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.2	6.4	11.3	37.3	1.6	0.9	10.2	37.9	9.3	38.5	8.4	39.0	7.3	39.7		
(5)			9.0	28.3	1.1	1.0	8.1	29.0	7.5	29.6	6.9	30.2	6.0	30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.2	28.5	28.6	28.0	27.5	24.6	21.8	21.4	22.5	25.6	27.9	29.2	29.4
	DBStd	3.37	1.71	1.45	1.46	1.66	2.15	2.52	2.01	2.05	1.67	1.52	1.47	1.52
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	6	0	0	0	0	0	3	2	1	0	0	0	0
	CDD10.0	5926	574	522	557	524	452	356	353	388	467	555	576	601
	CDD18.3	2889	316	289	299	274	194	108	97	130	217	297	326	343
	CDH23.3	30469	3797	3451	3346	2973	1509	539	434	835	1830	3258	4069	4428
	CDH26.7	11600	1532	1371	1240	1119	388	83	48	159	564	1236	1812	2047
Wind	WSAvg	3.0	3.0	2.4	2.4	2.5	3.1	3.4	3.2	3.0	3.2	3.2	3.3	3.0
Precipitation	PrecAvg	1285	316	331	293	65	11	6	2	1	2	16	56	170
	PrecMax	2114	881	1296	666	487	126	78	22	4	35	168	226	459
	PrecMin	583	65	41	31	0	0	0	0	0	0	0	0	21
	PrecStd	398	200	277	182	92	24	15	5	1	6	32	53	108
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	34.1	34.1	34.2	32.5	30.2	29.6	30.7	32.9	34.7	36.1	35.8
		MCWB	26.4	27.0	25.4	24.8	23.6	23.3	21.9	20.5	19.6	21.6	21.8	22.3
	2%	DB	33.0	33.0	33.1	32.9	30.9	28.7	27.9	29.4	31.4	33.2	34.3	34.1
		MCWB	26.7	27.0	25.4	24.9	23.5	22.1	20.6	20.1	21.0	22.8	23.2	24.4
	5%	DB	32.3	32.3	32.3	31.9	29.8	27.6	26.8	28.4	30.5	32.2	33.2	33.2
		MCWB	26.6	26.7	25.5	24.5	22.7	21.2	19.7	19.8	21.5	22.9	23.9	25.2
	10%	DB	31.6	31.7	31.4	31.2	28.9	26.3	25.8	27.4	29.6	31.4	32.3	32.3
		MCWB	26.5	26.6	25.4	24.3	22.3	20.3	19.1	19.4	21.4	23.1	24.3	25.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.6	29.8	28.6	27.6	26.0	24.9	23.1	23.4	25.0	27.4	28.5	27.5
		MCDB	31.9	32.7	31.0	30.6	29.8	28.3	27.3	27.7	28.6	30.5	32.1	32.0
	2%	WB	28.1	28.7	27.8	26.8	25.0	23.7	22.0	22.5	24.3	26.6	27.7	27.0
		MCDB	31.3	31.2	30.6	30.1	28.9	27.2	26.2	26.6	27.9	29.6	31.2	31.3
	5%	WB	27.5	28.1	27.2	26.2	24.2	22.6	21.2	21.9	23.8	25.9	26.7	26.6
		MCDB	30.8	30.6	30.0	29.4	28.1	26.1	25.1	25.8	27.3	29.1	30.3	30.8
	10%	WB	27.0	27.6	26.7	25.7	23.4	21.4	20.3	21.2	23.3	25.2	26.0	26.2
		MCDB	30.4	30.1	29.4	29.1	27.4	25.2	24.3	25.1	26.9	28.7	29.8	30.4
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.2	6.0	7.2	7.8	7.6	8.6	9.7	8.5	7.2	6.3	5.6
		MCDBR	5.6	6.2	7.4	8.2	8.8	9.1	9.8	10.6	9.0	8.1	7.5	6.5
		MCWBR	2.3	2.3	3.0	3.2	3.7	4.4	4.5	4.4	3.5	3.6	3.4	2.6
	5% WB	MCDBR	5.1	5.2	6.1	6.8	7.7	8.2	9.2	9.5	7.6	6.7	5.8	5.5
		MCWBR	2.5	2.3	3.2	3.2	3.7	4.2	4.5	4.5	3.1	3.0	3.0	2.4
Clear-Sky Solar Irradiance	taub	0.413	0.412	0.391	0.367	0.343	0.331	0.326	0.333	0.382	0.411	0.433	0.425	
	taud	2.532	2.534	2.589	2.630	2.640	2.630	2.633	2.592	2.459	2.407	2.396	2.461	
	Ebn at Noon	933	925	923	911	900	894	909	931	916	914	907	921	
	Edh at Noon	112	111	101	92	85	83	84	93	114	124	127	120	
All-Sky Solar Radiation	RadAvg	5.39	5.56	5.59	5.59	5.03	4.64	4.96	5.75	6.63	7.02	6.76	6.09	
	RadStd	0.69	0.82	0.66	0.48	0.27	0.27	0.12	0.13	0.19	0.35	0.44	0.67	